Barrier Methods (also known as interior-point methods)

是用于解决 constrained optimization problems 的一种方法 (一种迭代方法) Penalty method 也是迭代方法.

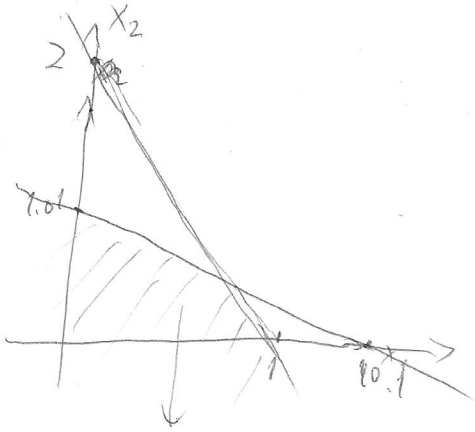
The core idea is to replace the inequality constraints with a smooth barrier function that penalizes points as they approach the boundary of the feasible region, while keeping iterates strictly inside the feasible set (the interior)

It enabled solutions of linear programming problems that were beyond the capacities of the simplex method. In contrast to the simplex method, it reaches a best solution by traversing the interior of the feasible region.

例子:

$$\max\ x_1+x_2$$
$$\text{s.t. } \underline{2px_1+x_2 \le p^2+1} \quad \forall p \in [0, 0.1, 0.2 \dots 1.0]$$

$$2x_1+x_2 \le 2$$
$$0.1x_1+x_2 \le 1.01$$



feasible region.

问题

(P_b) Penalty barrier.

$$\min\ f(x)$$
$$\text{s.t. } x \in \boxed{M[-g] = \{x \in \mathbb{R}^n \mid g_i(x) \ge 0,\ i \in I\}}$$

$M[-g]$ 自定义记号

Lagrangian: $L(x,\mu) = f(x) - \sum_{i \in I} \mu_i g_i(x)$

We define $M_0 := \{x \in \mathbb{R}^n \mid g_i(x) > 0,\ i \in I\}$

and assume $M_0 = M^\circ$

定义: Barrier-function. Let $B: M_0 \to \mathbb{R}$ be continuous. B is called a barrier-function 若

$$\lim_{x \to \bar{x} \in \partial M} B(x) = +\infty$$

M is a constraint set. $M[-g]$: 可行域

Penalty method: 允许超出可行域
Barrier method: 只允许在可行域里

相同之处: add ~~a penalty~~ a penalty like term to the objective function.